Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/02/2007
PI Name:	Jin, J.-P. M.D., Ph.D.		
Project Title:	Adaptation Of Myocardial Contractile Proteins In Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	jpjin@northwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	847.570.1960
Organization Name:	Evanston Northwestern Healthcare Research Institute		
PI Address 1:	2650 Ridge Ave		
PI Address 2:	Molecular Cardiology Division		
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60201-1718	**Congressional District:**	9
Comments:	Former affiliation: Case Western Reserve University; moved in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: NCE to 6/30/2008 per A. Chu (4/07) NOTE: NCE to 6/30/07 per A. Chu (7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK62G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In a rat tail-suspension model of simulated microgravity, we found an up-regulated proteolytic modification of cardiac troponin I (cTnI), an essential protein in the Ca2+-regulatory system of cardiac muscle. This novel structural regulation of cTnI removes the N-terminal 26-27 amino acids, including Ser23 and Ser24 that are substrates of cAMP-dependent protein kinase (PKA). Beta-adrenergic-regulated PKA phosphorylation of cTnI reduces the sensitivity of cardiac muscle to Ca2+. Therefore, the up-regulation of N-terminal truncated cTnI may provide a compensation for cardiac function against the microgravity-induced reduction of contractility. This research proposal describes ground based studies to investigate the role of the proteolytic regulation of cTnI in adaptation to microgravity for potential use in the prevention and treatment of cardiovascular deconditioning during and after space flight. We plan to study the functional changes in cTnI caused by the N-terminal truncation and the effect on cardiac muscle contraction in simulated microgravity. Conformation and function of the N-terminal truncated cTnI will be investigated in vitro and in the heart of transgenic mice. We shall examine the effect of the removal of PKA phosphorylation sites from cTnI on beta-adrenergic regulation during cardiac adaptation in simulated microgravity. We shall also investigate the regulatory mechanisms for this proteolytic modification of cTnI during adaptation to microgravity. We shall focus on the role of calpain that is a family of regulatory proteases and known to function in the degradation of troponin subunits. The role of PKA and PKC phosphorylation of cTnI in the regulation of sensitivity to calpain cleavage will be characterized. This research will provide new information for understanding cardiac adaptation in microgravity and protecting cardiac muscle function during and after long space flight. By investigating the molecular mechanism of cardiac muscle adaptation, our study will contribute to both space life science and human life quality on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our discovery that the N-terminal modification of cardiac troponin I enhances cardiac function has lead to an approach to the development of new drugs for the treatment of heart failure that is the most important death cause for cardiovascular diseases.
Task Progress:	Progress in the reporting period 1) Two research papers have been published to report results from this research project: Yu, Z.-B., Gao, F., Feng, H., and Jin, J.-P. (2007) Differential regulation of myofilament protein isoforms underlying the contractility changes in skeletal muscle unloading. Am. J. Physiol.:Cell Physiol., 292:C1192-203. Yu, Z.-B., and Jin, J.-P., (2007) Removing the regulatory N-terminal domain of cardiac troponin I diminishes incompatibility during bacterial expression. Archives Biochem. Biophys. 461:138-45. 2) A review article including work from this study is currently in press: Jin, J.-P., Zhang, Z., and Bautista, J.A. (2007) Isoform diversity, regulation and functional adaptations of troponin and calponin. Crit. Rev. in Eukar Gene Expr. In press. 3) A research paper from work in this study has been submitted for publication: Bradley K. McConnell, Zoran Popovic, Niladri Mal, James Bautista; Farhad Forudi, Raul Schwartzman, J.-P. Jin; Marc Penn; Meredith Bond; Targeted Disruption of PKA / AKAP Interaction: Effects on Cardiac Contractility, PKA Phosphorylation, and TnI Proteolysis in rat Hearts in vivo. 4) Mechanisms that regulate the proteolytic N-terminal truncation of cardiac troponin I: To investigate the physiological regulation of the cardiac troponin I N-terminal truncation and the role of PKA phosphorylation under beta-adrenergic regulation, we found increases of N-terminal truncation of cardiac troponin I in a Gs alpha conditional knockout mouse model. The disruption of Gs alpha signaling pathway blocks the beta-adrenergic signaling pathway. To test the hypothesis that the proteolytic deletion of the N-terminal domain of cardiac troponin I plays a role in compensating for the loss of beta-adrenergic stimulation in order to maintain cardiac function, we have thoroughly studied cardiac function of the Gs alpha conditional knockout mice. The results demonstrated a significantly decreased beta-adrenergic potential in the Gs alpha knockout hearts especially decreased relaxation that may be compensated by the function of N-terminal truncated cardiac troponin I. A manuscript is currently in preparation (Feng, H., M. Chen, Weinstein L.S., and Jin, J.-P. Heart failure phenotypes of myocardial Gsalpha knockout and compensation by proteolytic removal of the N-terminal domain of cardiac troponin I). 5) Presentations: The PI presented research results from this project in an invited lecture at the Sixth International Muscle Energetics Conference, Banff, Canada (07/2006). Title: Posttranslational modification of troponin in myocardial adaptation The PI presented research results from this project in an invited seminar at University of North Carolina at Chapel Hill (05/2007)
Bibliography Type:	Description: (Last Updated: 04/20/2009)
Articles in Peer-reviewed Journals	Yu ZB, Gao F, Feng HZ, Jin JP. "Differential regulation of myofilament protein isoforms underlying the contractility changes in skeletal muscle unloading." Am J Physiol Cell Physiol. 2007 Mar;292(3):C1192-203. PMID: 17108008 , Mar-2007
Articles in Peer-reviewed Journals	Yu ZB, Jin JP. "Removing the regulatory N-terminal domain of cardiac troponin I diminishes incompatibility during bacterial expression. " Arch Biochem Biophys. 2007 May 1;461(1):138-45. PMID: 17303066 , May-2007
Articles in Peer-reviewed Journals	Jin J-P, Zhang Z, Bautista JA. "Isoform diversity, regulation and functional adaptations of troponin and calponin." Crit Rev Eukaryot Gene Expr. in press, June 2007. , Jun-2007